

You will be well-prepared to navigate the fascinating world of Python programming, create eye-catching projects, and start a rewarding career in software development by making use of these materials and modules.

Working with APIs

APIs (application programming interfaces) facilitate communication and interaction between different software systems, enabling seamless integration and data exchange. They have become integral to enabling communication between web applications and services. Web APIs expose specific functionalities or data endpoints over standard protocols such as HTTP, allowing developers to access and manipulate resources remotely. This has fuelled the growth of the web ecosystem, giving rise to platforms and services that leverage APIs for various purposes, including social media integration, e-commerce transactions, and cloud computing. Today, APIs are ubiquitous across the digital landscape, powering a vast array of applications, platforms, and devices. From social media APIs like Facebook Graph API to payment gateways like Stripe API, APIs have become the building blocks of interconnected digital ecosystems. The emphasis on interoperability and standardisation has led to the proliferation of RESTful APIs, which adhere to the principles of simplicity, scalability, and uniformity.

Types of API

APIs can be sorted based on their functions, uses, and methods of exposing data or services.

Web interaction based APIs

RESTful APIs: Representational State Transfer APIs use HTTP requests to perform CRUD (create, read, update, delete) operations. These APIs typically send back data in JSON or XML format.

SOAP APIs: Simple Object Access Protocol APIs use XML for message exchange over HTTP, SMTP, or TCP/IP. These APIs use XML Schema Definition (XSD) for message structure.

GraphQL APIs: A query language-based API that enables clients to request only the needed data, allowing more flexibility and efficiency in data retrieval.

JSON-RPC and XML-RPC APIs: Remote Procedure Call protocols allow client and server communication using JSON or XML messages.

Operating system-based APIs

Windows API: This API manages resources and interacts with the Windows operating system.

POSIX API: Portable Operating System Interface for UNIX-like operating systems provides standard functionalities across different UNIX and UNIX-like systems.

Library APIs

Standard libraries: These are bundled libraries provided by programming languages for everyday tasks like file I/O, networking, file transfer, manipulation, etc.

Third-party libraries: These API libraries are developed by third-party developers such as Stripe to make the development easy to access and extend the functionality of programming languages or frameworks.

Hardware interaction APIs

Graphics APIs: DirectX and OpenGL render graphics on computer screens.

Device APIs: Help to access hardware devices such as cameras, sensors, and peripherals.

Database interaction APIs

SQL APIs: Application interfaces for interacting with relational databases using SQL (Structured Query Language).

NoSQL APIs: NoSQL databases, such as MongoDB, Cassandra, etc, allow APIs to interact with production stack-driven applications.

Cloud interaction APIs

Storage APIs: Allow interaction with cloud storage services like Amazon S3, Google Cloud Storage, Azure Blob Storage, etc.

Compute APIs: Enable management and provisioning of virtual machines, containers, and serverless functions in cloud platforms like AWS, Azure, Google Cloud, etc.

AI/ML APIs: Provide access to machine learning and artificial intelligence services such as natural language processing, image recognition, etc.

Social media APIs: Facebook Graph API, Twitter API, Instagram API, etc, provide access to social media platforms' functionalities for integrating third-party applications.

Enterprise APIs

Enterprise Service Bus (ESB) APIs: Facilitate communication between different enterprise applications and systems.

CRM APIs: Enable developers within an organisation to access customer relationship management systems like Salesforce, SAP, etc, and allow them to integrate for further application development.

ERP APIs: Allow integration with enterprise resource planning systems for managing business processes.

Playing around with an API

Let's make it simpler to understand what APIs do. Basically, they act as a mediator between a channel and client. Let's say a backend client wants to inform the frontend that the task is complete. Here comes the API's role — this little guy will deliver our message to the frontend client.